



Hah Min Lew

Ph.D. student · Korea University

[hahminlew.github.io](https://github.com/hahminlew)

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Biography

Ph.D. student at the Visual & General Intelligence (VGI) Lab., Korea University, supervised by Prof. Gyeong-Moon Park. Current research focus lies in Vision-Language-Action (VLA) models, Vision-Language-Navigation (VLN) models, and multimodal generative models. Recipient of the *Presidential Science Scholarship* funded by the Ministry of Science and ICT.

Prior to academia, professional experience includes over three years as an AI Researcher at Klleon AI Research. Key contributions involved photo-realistic 3D human generation, enabling lifelike avatar movements and lip-sync. Holds both M.S. and B.E. degrees from DGIST, with a strong foundation in Deep Learning and Signal Processing, advised by Prof. Jae Youn Hwang.

Education

Korea University (Advisor: Prof. Gyeong-Moon Park)

Ph.D. in Artificial Intelligence

Seoul, South Korea

Mar. 2026 - present

DGIST (Advisor: Prof. Jae Youn Hwang)

M.S. in Electrical Engineering and Computer Science

Daegu, South Korea

Mar. 2019 - Aug. 2022

DGIST (Daegu Gyeongbuk Institute of Science and Technology)

B.E. in School of Undergraduate Studies

Daegu, South Korea

Mar. 2014 - Feb. 2019

Experience

Klleon AI Research

AI Researcher

Seoul, South Korea

Aug. 2022 - Feb. 2026

- Accelerated inference using a training-free diffusion sampler (4.11x).
- Developed audio-driven 3DMM generation for virtual avatars with natural human-like expressions and movements. (LVE 27.5%, FDD 28.9%, MEE 27.1%, CE 24.1%, Diversity 17.7%)
- Developed a photorealistic head rendering model using Gaussian Splatting, outperforming 5 state-of-the-art models. (MSE 59.96%, PSNR 4.41dB, SSIM 3.85%, LPIPS 38.16%)
- Developed a Head Swap AI model with 212.7x speedup, reducing annual GPU costs by 99.53% (from \$2.2M to \$10.5K).
- Built large-scale multimodal data pipelines (9.41M+ frames from in-the-wild videos).
- Integrated external APIs (NVIDIA Omniverse Audio2Face, OpenAI ChatGPT, TTS) into a streaming avatar system, achieving 25-28 FPS performance.

Multimodal Biomedical Imaging and System Lab, DGIST

Graduate Researcher

Daegu, South Korea

Mar. 2019 - Aug. 2022

- Achievements: 6 SCIE publications, 7 international conferences, 9 projects, 4 patents, and 2 awards.
- Designed machine learning-based anomaly detection systems and generative models for biomedical imaging, focusing on multimodal, class-imbalance, and multi-task learning.
- Developed hardware-software integrated systems for application-specific use cases.
- Collaborative research with medical doctors from hospitals, including SNUH, SNUDH, and Yonsei Severance.

Publications

[P1] FacePlex: Full-Duplex Joint Speech-Facial Motion Generation for Conversational Avatars

H. Lim*, J.-H. Lee*, **Hah Min Lew***, J.-S. Kang, and G.-M. Park. *arXiv* 2026

First Author

Jun. 2026

[C5] AudioAvatar: Personalized Audio-driven Whole-body Talking Avatars

S. Lee, S. Moon, **Hah Min Lew**, J.-S. Kang, and G.-M. Park. *CVPR* 2026

Co-Author

Jun. 2026

[C4] SMAGA: Secondary Motion-Aware 3D Clothed Gaussian Avatars from Monocular Videos

S. Lee, S. Moon, **Hah Min Lew**, J.-S. Kang, and G.-M. Park. *ICLR* 2026

Co-Author

Apr. 2026

[C3] GeoAvatar: Adaptive Geometrical Gaussian Splatting for 3D Head Avatar

S. Moon*, **Hah Min Lew***, S. Lee, J.-S. Kang, and G.-M. Park. *ICCV* 2025

First Author

Oct. 2025

[C2] Towards High-fidelity Head Blending with Chroma Keying for Industrial Applications

Hah Min Lew*, S.-M. Yoo*, H. Kang*, and G.-M. Park. *WACV* 2025

First Author

Mar. 2025

[C1] CSS-Net: Classification and Substitution for Segmentation of Rotator Cuff Tear

K. Lee, **Hah Min Lew**, M. H. Lee, M. Kang, J. Kim, and J. Y. Hwang. *ACCV* 2022

Co-Author

Dec. 2022

[J6] Multi-task and Few-shot Learning-based Fully Automatic Deep Learning Platform for Mobile Diagnosis of Skin Diseases	Co-Author
K. Lee, T. C. Cavalcanti, S. Kim, Hah Min Lew , D. H. Suh, D. H. Lee, and J. Y. Hwang. <i>IEEE JBHI</i> (IF: 7.021)	Jan. 2023
[J5] Deep Learning-based Framework for Fast and Accurate Acoustic Hologram Generation	Co-Author
M. H. Lee, Hah Min Lew , S. Youn, T. Kim, and J. Y. Hwang. <i>IEEE TUFFC</i> (IF: 3.267)	Dec. 2022
[J4] Speckle Reduction via Deep Content-Aware Image Prior for Precise Breast Tumor Segmentation in an Ultrasound Image	Co-Author
H. Lee, M. H. Lee, S. Youn, K. Lee, Hah Min Lew , and J. Y. Hwang. <i>IEEE TUFFC</i> (IF: 3.267)	Sep. 2022
[J3] Intelligent Smartphone-based Multimode Imaging Otoscope for the Mobile Diagnosis of Otitis Media	Co-Author
T. C. Cavalcanti, Hah Min Lew , K. Lee, S. Lee, M. K. Park, and J. Y. Hwang. <i>Biomedical Optics Express</i> (IF: 3.562)	Dec. 2021
[J2] Ultrasonic Blood Flowmeter with a Novel Xero Algorithm for a Mechanical Circulatory Support System	First Author
Hah Min Lew , H. Shin, M. H. Lee, S. Youn, H. C. Kim, and J. Y. Hwang. <i>Ultrasonics</i> (IF: 4.062)	Aug. 2021
[J1] Forward-Looking Multimodal Endoscopic System Based on Optical Multispectral and High-Frequency Ultrasound Imaging Techniques for Tumor Detection	Co-Author
J. Kim, Hah Min Lew , J. Kim, S. Youn, H. A. Faruque, A. N. Seo, S. Y. Park, J. H. Chang, E. Kim, and J. Y. Hwang. <i>IEEE TMI</i> (IF: 11.037)	Feb. 2021

Patents

COMPUTING DEVICE FOR HEAD SWAPPING	Application
Hah Min Lew , H. Kang, S.-M.Yoo, G.-M.Park (WO2025042068A1, KR1020230154188A)	Feb. 2025
MOBILE OTOSCOPE SYSTEM	Patent
J. Y. Hwang, T. C. Cavalcanti, Hah Min Lew (KR102648059B1)	Mar. 2024
ULTRASONIC BLOOD FLOW MEASURING APPARATUS AND METHOD THEREOF	Patent
J. Y. Hwang, Hah Min Lew , H. C. Kim (KR102514633B1)	Mar. 2023
BLADDER MONITORING APPARATUS AND METHOD FOR CONTROLLING BLADDER MONITORING APPARATUS	Patent
J. Y. Hwang, M. H. Lee, Hah Min Lew (KR102460829B1)	Oct. 2022
THREE-DIMENSIONAL DIAGNOSTIC SYSTEM	Patent
J. Y. Hwang, J. Kim, Hah Min Lew (KR102379481B1)	Mar. 2022

Awards & Scholarships

Presidential Science Scholarship for Graduate Students (3rd Cohort)	South Korea
Ministry of Science and ICT	Mar. 2026 – Present
Outstanding Poster Award	Daegu, South Korea
2021 Student Conference at DGIST	2021
Best Paper Award	Remote, South Korea
2021 Spring Conference at KOSOMBE	2021
Best Project Award	Daegu, South Korea
2016 Undergraduate Group Research Project (UGRP) Program at DGIST	2017
Full Government Scholarships	Daegu, South Korea
Full tuition exemptions and school expenses support	2014-2022

Skills

Programming	Python, Bash, MATLAB, C, Java
Frameworks	PyTorch, TensorFlow, Keras, Scikit-learn, Pytorch3D
Tools	Docker, Containerd, Git, Node.js, HTML, CSS, JavaScript
Languages	Korean, English